

EXECUTIVE SUMMARY

Product Line Profitability & Margin Performance Analysis — Nassau Candy Distributor

Prepared For: Unified Mentor

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Purpose:

This report summarizes a data-driven profitability analysis conducted for Nassau Candy Distributor, a specialty manufacturer, importer, and national wholesale distributor operating across six regional distribution centers. The study applies established cost-accounting and business-intelligence methods to determine which products, divisions, and regions genuinely drive profit, as distinct from revenue alone, and is submitted as a reference case for [supply-chain resilience / MSME analytics adoption / distribution-sector policy] purposes.

Background:

The analysis covers 8,549 order line items across 10,194 verified transactions, spanning 15 products in 3 divisions (Chocolate, Sugar, Other), sold across 4 U.S. regions and 59 states. Data was independently validated, with Gross Profit confirmed to equal Sales minus Cost across all records, before four analytical lenses were applied: product-level ranking, division comparison, profit-concentration (Pareto) analysis, and cost-structure diagnostics.

Key Findings:

Metric	Finding
Overall Gross Margin	65.9%
Revenue from Top 5 Products (of 15)	92.9% of revenue, 95.1% of profit
Highest-Risk Product	Kazookles — 7.7% margin, 92.3% cost ratio
Weakest Division	“Other” — 6.8% of revenue, only 4.6% of profit
Geographic Concentration	16 of 59 states drive 80% of revenue

- **Profit concentration:** Just 5 of 15 products, all Chocolate Wonka Bar variants, generate over 92% of revenue and 95% of profit, well beyond the classic 80/20 threshold.
- **Margin risk identified:** One product (Kazookles) consumes 92.3% of its own sale price in cost, actively eroding value at meaningful volume (371 units, 94 orders), the single clearest actionable finding in the analysis.
- **Division imbalance:** The Chocolate division over-delivers profit relative to revenue (+2.2 points); the “Other” division under-delivers (–2.2 points), traced almost entirely to one underpriced product.
- **Structural, not seasonal:** Per-product margins are stable year-round, meaning margin problems stem from fixed pricing and costing decisions rather than promotional cycles, a simpler, one-time fix rather than an ongoing discipline issue.
- **Geographic footprint:** Revenue is moderately concentrated by geography (California, New York, and Texas lead), a normal pattern but relevant to logistics and demand planning.

Public Interest:

- **Supply-chain resilience:** The finding that roughly 93% of profit depends on five SKUs illustrates a concentration-risk pattern common across mid-size distributors, relevant to any public program monitoring supply-chain fragility in food and consumables distribution.
- **MSME cost-accounting practice:** The shift demonstrated here, from Volume-Based Costing to Activity-Based Costing, is a transferable model for advisory programs helping smaller distributors move beyond top-line revenue tracking.
- **Regional economic activity:** Uneven regional performance (the Gulf region at 15.7% of revenue versus the Pacific region at 32.7%) offers a data point for regional trade or market-development initiatives.

Risk Considerations:

- Any disruption to Wonka Bar manufacturing, sourcing, or demand would materially affect the entire business, a textbook single-point-of-failure concentration.
- Kazookles and several Sugar-division long-tail products (SweeTARTS, Nerds, Fun Dip) sit in high-risk margin territory, warranting cost renegotiation or discontinuation review.
- The Gulf region's underperformance relative to comparable regions suggests unrealized market opportunity rather than a structural problem.

Recommendations:

1. Renegotiate or restructure the cost base of the highest-risk product; target a minimum 30% gross margin within two quarters, with discontinuation as a fallback.
2. Reduce over-dependence on the top-performing division by growing the combined share of the other two divisions to at least 15% of revenue within 18 months.
3. Reprice or renegotiate costs for products sitting at the margin-risk threshold (around 50% margin).

4. Rationalize low-value, long-tail SKUs that contribute negligible profit relative to catalog and logistics overhead.
5. Investigate expansion opportunities in the underperforming Gulf region.
6. Institutionalize monthly dashboard review of cost-vs-margin diagnostics as a standing management practice.

Resource Implications:

The analytical framework relies on standard business-intelligence tools (Python, Streamlit, Plotly) and existing transactional data, requiring no new data infrastructure, a low-cost, replicable model for similar distribution-sector analysis.